

Fun with Words

(Word Embeddings)

- 1. Basics - Words, Tokens, Embeddings**
- 2. Similar, Different & Plots**
- 3. Having Fun (Exploring)**
- 4. Useful?**

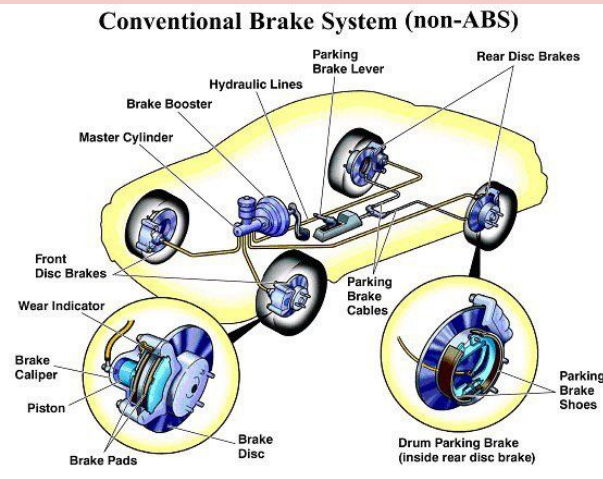
Topics

1. Pre-trained - Word2Vec, GloVe
2. King - man + woman
3. Python - GenSim
4. Comparing “Apples and Oranges”
5. Wicket keeper: Cricket and ??: Soccer
 - a. Do embeddings understand analogies?
6. Is a Surgeon closer to Man or to Woman?



Is it necessary to know
about
transmission/calipers/hydraulics

To drive an automatic car?
To be a stick shift driver?
To be a race car driver?



Embeddings



Why do we write computer
programs in
Python or C++ or Java?

Why not in plain English?

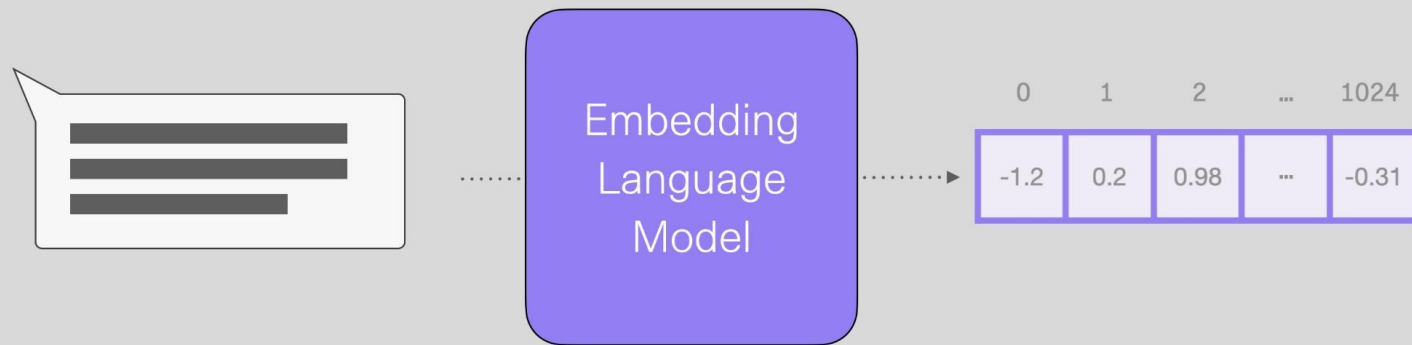
Embeddings

In short, a word embedding represents the word's semantic and syntactic meaning, often within a specific context. These embeddings can be obtained as part of training the Machine Learning model, or by means of a separate training procedure. Usually, word embeddings consist of between tens and thousands of variables, per word that is.

Embeddings

Embeddings are continuous vector representations of words or tokens that capture their semantic meanings in a high-dimensional space. They allow the model to convert discrete tokens into a format that can be processed by the neural network. In LLMs, embeddings are learned during the training process, and the resulting vector representations can capture complex relationships between words, such as synonyms or analogies.

Embeddings are numeric representations of text, capturing its meaning



co:here

3- Computers numerically represent text in unicode, which is excellent for string matching and to power tools like RegEx and Grep. Language processing needs another representation scheme. Text embeddings are numeric vectors. They enable software to compare text meaning.

Text documents
(or sentences)



Embed & group
by similarity



Cluster 1



Cluster 2

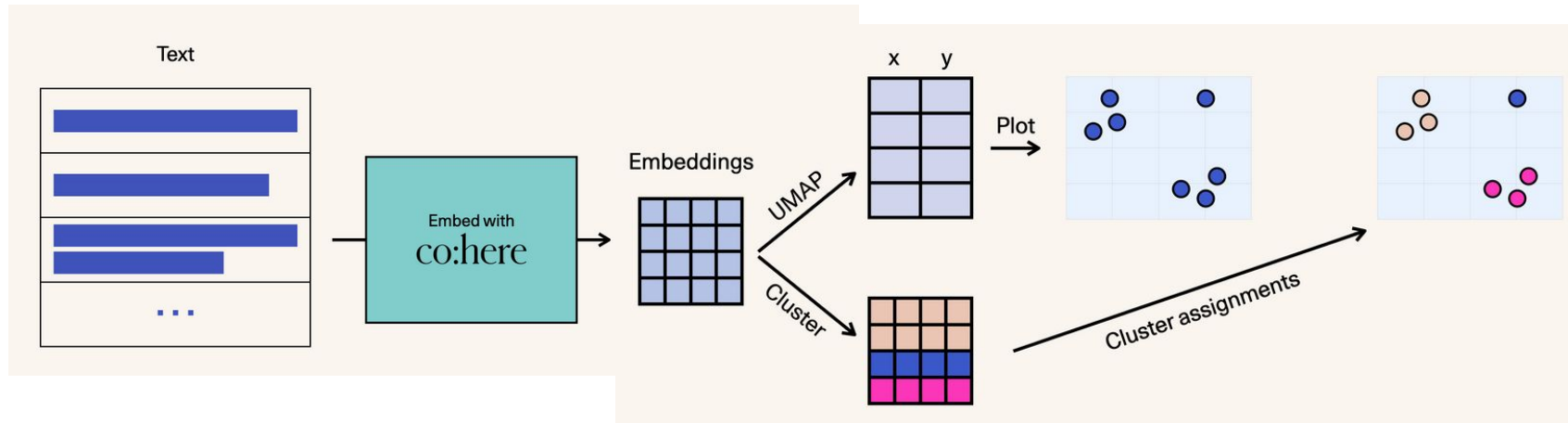


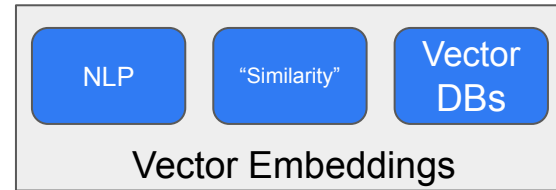
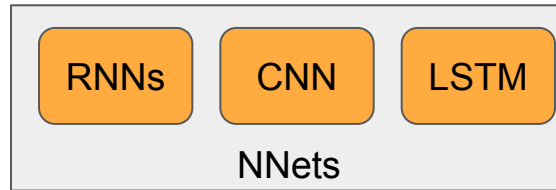
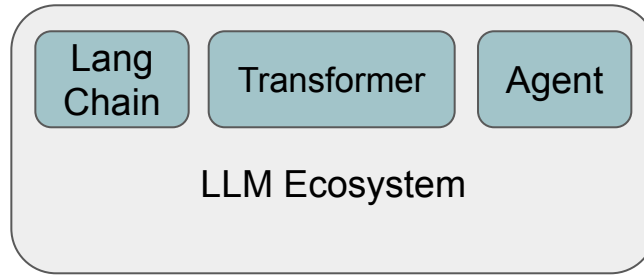
Cluster 3



From Sentence to Numeric Embeddings

Take an article's title (10-15 words) and create a 1024 number embedding





Nuts and Bolts

Tokenization: The process of breaking down text into smaller units called tokens, typically words or subwords, for processing by natural language models.

Dot Product: A mathematical operation that takes two equal-length sequences of numbers and returns a single number by multiplying corresponding entries and summing the results.

Cosine Similarity: A measure of similarity between two non-zero vectors calculated by the cosine of the angle between them, often used to compare document or embedding similarity.

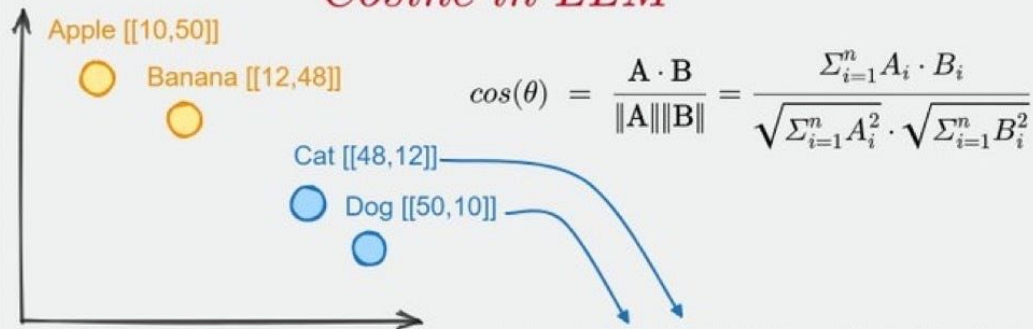
Examples

Tokenization: The sentence "I love machine learning" might be tokenized as: ["I", "love", "machine", "learning"] Or in subword tokenization: ["I", "love", "machine", "learn", "###ing"]

Cosine Similarity: If document vectors are $A = [1, 1, 1]$ and $B = [2, 2, 2]$ Cosine similarity = 1 (perfectly similar), as they point in the same direction

Similarity

Cosine in LLM

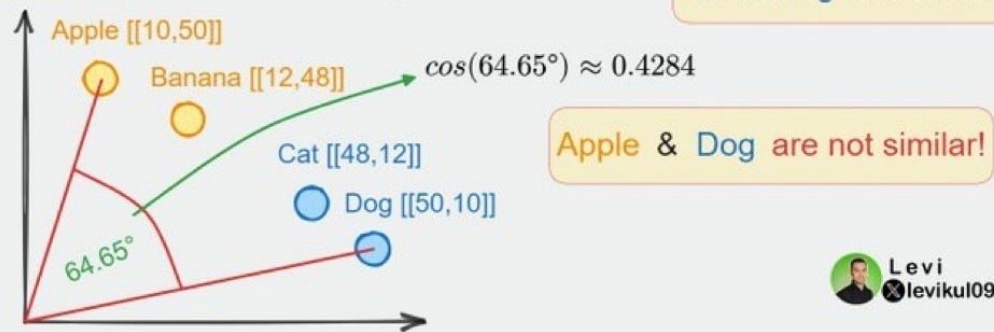
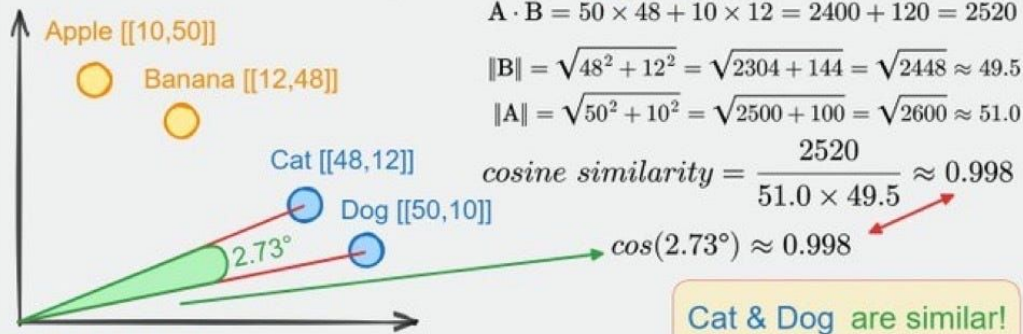


$$\mathbf{A} \cdot \mathbf{B} = 50 \times 48 + 10 \times 12 = 2400 + 120 = 2520$$

$$\|\mathbf{B}\| = \sqrt{48^2 + 12^2} = \sqrt{2304 + 144} = \sqrt{2448} \approx 49.5$$

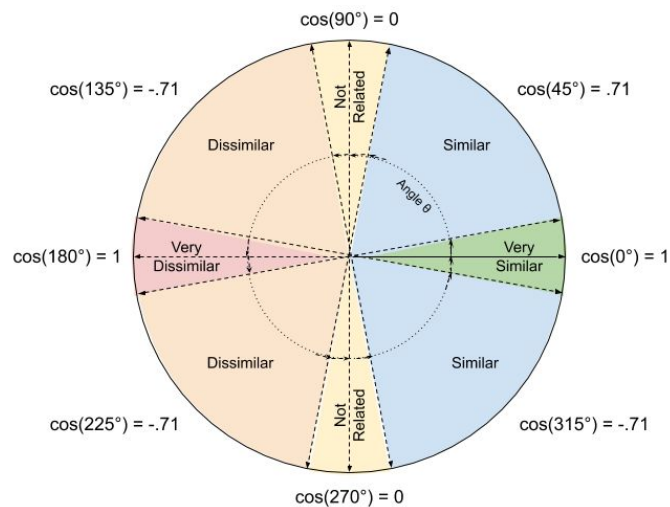
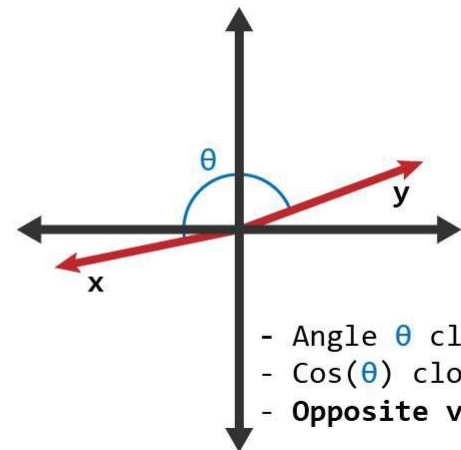
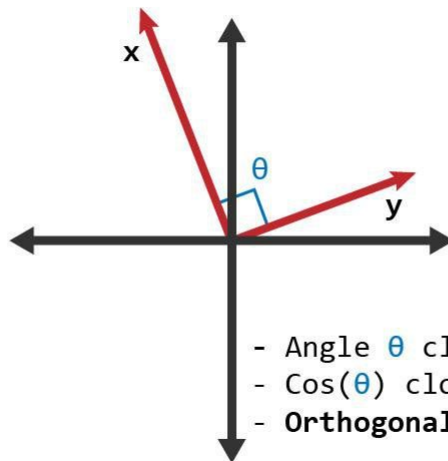
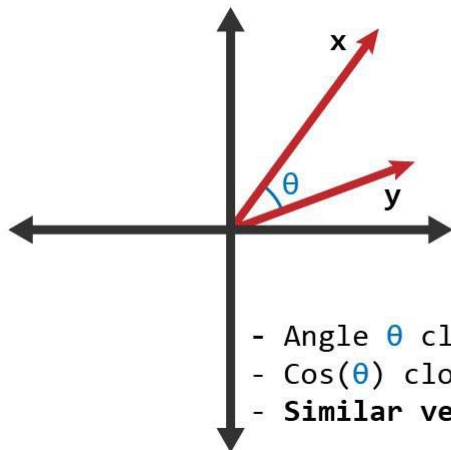
$$\|\mathbf{A}\| = \sqrt{50^2 + 10^2} = \sqrt{2500 + 100} = \sqrt{2600} \approx 51.0$$

$$\text{cosine similarity} = \frac{2520}{51.0 \times 49.5} \approx 0.998$$



Cat and Dog are similar
Apple and Banana are similar

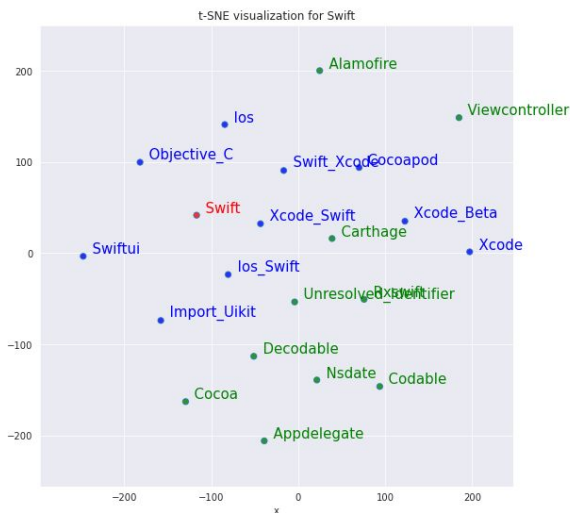
Apple and Dog are not



Gensim Word2Vec usage with t-SNE plot .

Kaggle Notebook

<https://www.kaggle.com/code/achintyatripathi/gensim-word2vec-usage-with-t-sne-plot>



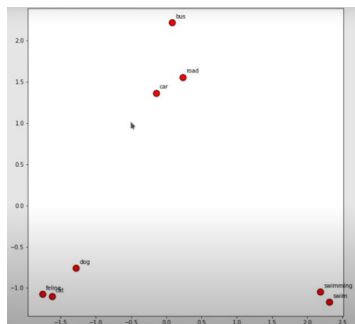
Visually Exploring Embeddings

<https://huggingface.co/docs/hub/datasets-embedding-atlas>

<https://www.youtube.com/watch?v=5MaWmXwxFNQ>

Colab:

https://colab.research.google.com/github/futuremojo/nlp-demystified/blob/main/notebooks/nlpdemystified_word_vectors.ipynb



```
In [32]: fasttext.most_similar_cosmul(positive=['restaurant','cocktail'])
```

```
Out[32]: [('bar-restaurant', 0.9216465950012207),  
          ('restaurant-bar', 0.9107573628425598),  
          ('cocktail-making', 0.8984322547912598),  
          ('mixology', 0.8981987833976746),  
          ('wine-bar', 0.8936376571655273),  
          ('nightclub', 0.889917254447937),  
          ('brewpub', 0.8862351775169373),  
          ('gastro-pub', 0.8823451995849609),  
          ('cafe-bar', 0.8814408779144287),  
          ('bar-cafe', 0.8773310116386414)]
```

```
wv = api.load('word2vec-google-news-300')
```

```
[=====] 100%
```

```
glove = api.load("glove-twitter-50")
```

```
[=====] 100%
```

```
fasttext = api.load("fasttext-wiki-news-subwords-300")
```

```
[=====] 100%
```

```
!pip install --upgrade gensim
```

...

```
import gensim.downloader as api
```

```
info = api.info()  
for model_name, model_data in sorted(info['models'].items()):  
    print(  
        '%s (%d records): %s' % (  
            model_name,  
            model_data.get('num_records', -1),  
            model_data['description'][:40] + '...',  
        )  
    )
```

```
__testing_word2vec-matrix-synopsis (-1 records): [THIS IS ONLY  
...  
conceptnet-numberbatch-17-06-300 (1917247 records): ConceptNet  
e...  
fasttext-wiki-news-subwords-300 (999999 records): 1 million wo  
e...  
glove-twitter-100 (1193514 records): Pre-trained vectors based  
glove-twitter-200 (1193514 records): Pre-trained vectors based  
glove-twitter-25 (1193514 records): Pre-trained vectors based
```